

WEEK 5 REPORT

Healthcare Data Cleaning & Data Modeling

(Power BI Desktop)

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1. Introduction

This week I worked on a healthcare dataset instead of the stock data. The raw data had patient details, admission details and billing details all mixed in one big table, so my job was to clean it up and split it into a proper data model before it can be used for reporting.

2. Objectives

- Import and inspect patient records, admission types, and billing fields in Power Query.
- Handle null values in diagnostic codes and standardize categorical text attributes.
- Create derived conditional columns, for example Age Group buckets: 0-18, 19-35, 36-60, 60+.
- Structure dimension tables (Dim_Patient, Dim_Admission) and a fact table (Fact_Billing).

3. About the Dataset

The dataset is called `healthcare_dataset`, and each row is one patient admission record. It has columns like Name, Age, Gender, Blood Type, Medical Condition, Date of Admission, Doctor, Hospital, Insurance Provider, Billing Amount, Room Number, Admission Type, Discharge Date, Medication and Test Results.

4. Steps

Step 1: Import and inspect the data

I loaded the healthcare_dataset file into Power Query. The first row was used as headers, and Power Query automatically set the data types for each column, for example Age as a whole number and Date of Admission as a date. All the columns showed 100% valid values with no errors.

File Home Insert Modeling View Optimize Help

File Home Transform Add Columns View Tools Help

Close & Apply * New Recent Enter Data Sources * Sources * Data Close New Query

Data sources Parameters * Export query Visuals Refresh Review * Query

Properties Advanced Editor Manage * Choose Remove Columns * Columns * Manage Columns

Keep Remove Columns * Reduce Rows Sort

Data type Text * Split Group By * Use First Row as Headers * L₁ Replace Values * Transform

Merge Queries * Append Queries * Combine Files * Combine

Queries [1] < X ✓ **Table.TransformColumnTypes(*Promoted Headers, ({Name, type text}, {"Age", Int64.Type}, {"Gender", type text}),**

healthcare_dataset

	Age	Gender	Blood Type	Medical Condition
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1 Bobby Jackson	30	Male	B-	Cancer
2 Leslie Terry	62	Male	A+	Obesity
3 Diament SMITH	76	Female	A-	Obesity
4 andrew watts	28	Female	O+	Diabetes
5 adrienne DEE	43	Female	AB-	Cancer
6 EMERY JOHNSON	36	Male	A+	Asthma
7 melinda EDWARDS	23	Female	A-	Diabetes
8 CHRISTINA MARTINEZ	21	Female	AB+	Cancer
9 JACIMILE aGullati	62	Male	AB+	Asthma
10 CHRISTOPHER BERG	58	Female	AB-	Cancer
11 michelle daniels	72	Male	O+	Cancer
12 aaron MARTINEZ	38	Female	A-	Hypertension
13 conOR HANCOIN	75	Female	A+	Diabetes
14 rochelle Baker	61	Female	AB+	AB+
15 BROCKE brady	48	Female	AB-	Cancer
16 MRS. nashali gAMble	46	Female	AB-	Obesity
17 haley perkins	63	Female	A+	Arthritis

15 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 08:49

PROPERTIES

Name healthcare_dataset

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

After clicking OK, the new Age Group column showed the correct bucket for every patient, for example 19-35 or 60+.

The screenshot shows a data table with the following columns: Billing Amount, Room Number, Admission Type, Discharge Date, Medication, Test Results, and Age Group. The Age Group column is the result of a formula: "Table.AddColumn('Changed Type1', 'Age Group', each if [Age] <= 18 then '0-18'". The table contains 25 rows of patient data, including fields like Name, Age, Gender, Blood Type, Medical Condition, and Age Group.

Step 4: Build the dimension and fact tables

The raw table had patient details, admission details and billing details all mixed together, so I split it into separate tables. I used Choose Columns to keep only the patient related fields (Name, Age, Gender, Blood Type, Medical Condition, Age Group) for a table called Dim_Patient.

The screenshot shows the 'Choose Columns' dialog box in a data tool. The dialog has a search bar and a list of columns to choose from. The columns listed are: Name, Age, Gender, Blood Type, Medical Condition, Date of Admission, Doctor, Hospital, Insurance Provider, Billing Amount, Room Number, Admission Type, Discharge Date, Medication, Test Results, and Age Group. The 'Choose Columns' dialog is open, and the 'Dim_Patient' table is selected in the left pane. The 'Choose Columns' dialog shows the following columns selected: Name, Age, Gender, Blood Type, Medical Condition, and Age Group.

In the same way, I created a **Dim_Admission** table by removing the unnecessary columns and keeping only the admission-related fields such as **Admission ID**, **Date of Admission**, **Admission Type**, and **Discharge Date**.

And a Fact_Billing table by removing the columns that were not needed, keeping only the billing and admission related fields like Admission ID, Billing Amount, Date of Admission, Hospital and Insurance Provider.

Formula Bar: `Table.SelectColumns(Source,{\"Name\", \"Medical Condition\", \"Date of Admission\", \"Hospital\", \"Insurance Provider\"})`

	Name	Medical Condition	Date of Admission	Hospital	Insurance Provider
1	Bobby Jackson	Cancer	31-01-2024	Sons of	
2	Leslie Terry	Obesity	20-08-2019	Kim In	
3	Danny Smith	Obesity	22-09-2022	Cook	
4	Andrew Watts	Diabetes	18-11-2020	Herna	
5	Adrienne Bell	Cancer	19-09-2022	White	
6	Emily Johnson	Asthma	20-12-2023	Nunez	
7	Edward Edwards	Diabetes	03-11-2020	Group	
8	Christina Martinez	Cancer	28-12-2021	Powell	
9	Jasmine Aguilar	Asthma	01-07-2020	Sons of	
10	Christopher Berg	Cancer	23-05-2021	Padilla	
11	Michelle Daniels	Cancer	19-04-2020	Schae	
12	Aaron Martinez	Hypertension	13-08-2023	Lyons	
13	Connor Hansen	Diabetes	12-12-2019	Powell	
14	Robert Bauer	Asthma	22-05-2020	Rivera	
15	Brock Brady	Cancer	08-10-2021	Morris	
16	Ms. Natalie Gamble	Obesity	01-01-2023	Cline	
17	Haley Perkins	Arthritis	23-06-2020	Cerv	
18	Mrs. Jamie Campbell	Obesity	08-03-2020	Torre	
19	Luke Burgess	Hypertension	04-03-2021	Houst	
20	Daniel Schmidt	Asthma	15-11-2022	Hamm	
21	Timothy Burns	Asthma	28-06-2023	Jones	
22	Christopher Bright	Asthma	21-01-2020	Willia	
23	Kathryn Stewart	Arthritis	12-05-2022	Clark	
24	Dr. Ellen Thompson	Asthma	02-08-2021	and S	
25	Paul Henderson	Hypertension	15-05-2020	Willso	
26	Peter Fitzgerald	Obesity	15-05-2020	Garne	
27	Cathy Small	Asthma	23-12-2023	Brown	

Step 5: Check the tables in Model view

After loading, the Model view showed four tables: Dim_Patient, Dim_Admission, Fact_Billing, and the original healthcare_dataset. At this point none of them were connected to each other yet.

Model view showing four tables:

- Dim_Patient**
 - Age Group
 - Blood Type
 - Gender
 - Medical Condition
 - Name
- Dim_Admission**
 - Admission Type
 - Date of Admission
 - Discharge Date
- Fact_Billing**
 - Admission Type
 - Billing Amount
 - Date of Admission
 - Hospital
 - Insurance Provider
 - Medical Condition
 - Name
- healthcare_dataset**
 - Admission Type
 - Age Group
 - Billing Amount
 - Blood Type
 - Date of Admission
 - Discharge Date
 - Doctor
 - Gender

Step 6: Create the relationships

I connected the tables using a Patient Key column: Dim_Patient to Fact_Billing as One to one, with cross-filter direction set to Both and the relationship marked active. I also connected Dim_Admission to the fact and raw tables the same way.

The screenshot displays the Power BI Desktop interface. The top pane shows the 'Model' view with four tables: Dim_Patient, Dim_Admission, Fact_Billing, and healthcare_dataset. Dim_Patient is connected to Fact_Billing and healthcare_dataset. Dim_Admission is connected to Fact_Billing and healthcare_dataset. The bottom pane shows the 'Edit relationship' dialog box.

Edit relationship
Select tables and columns that are related.

From table: Dim_Patient

Age	Age Group	Blood Type	Gender	Medical Cond...	Name	Patient Ke
72	60+	O+	Male	Cancer	Michelle Dani...	10
65	60+	O+	Male	Cancer	Lynn Martinez	45
78	60+	O+	Male	Cancer	Michael Wolfe	845

To table: Fact_Billing

Admission ID	Billing Amount	Date of Admi...	Hospital	Insurance Pro...	Patient Key
1	33643.327286...	20 August 2019	Kim Inc	Medicare	1
3	37909.782409...	18 November...	Hernandez R...	Medicare	3
6	19580.872344...	03 November...	Group Middle...	Medicare	6

Cardinality: One to one (1:1)

Cross-filter direction: Both

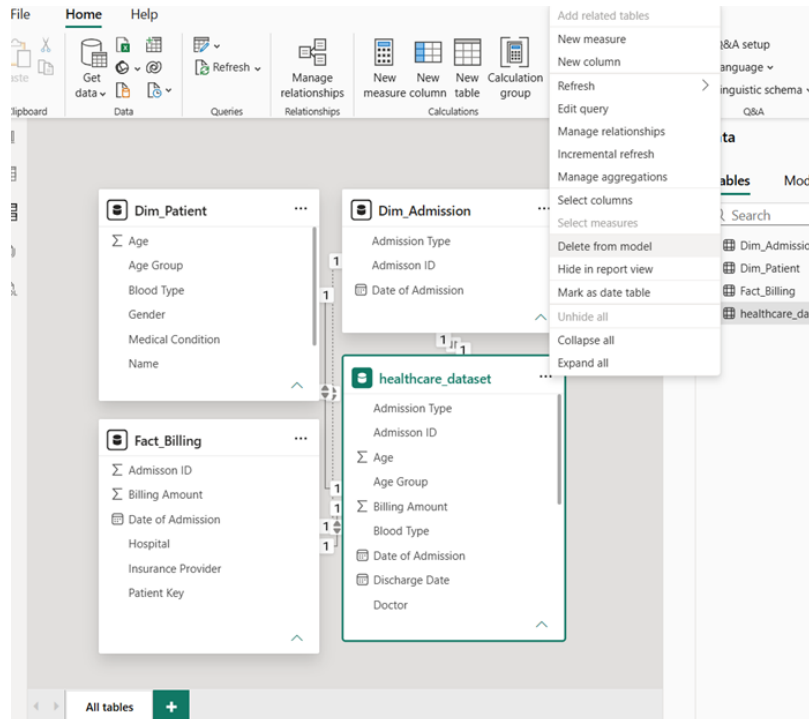
☒ Make this relationship active

☐ Assume referential integrity

Buttons: Save, Cancel

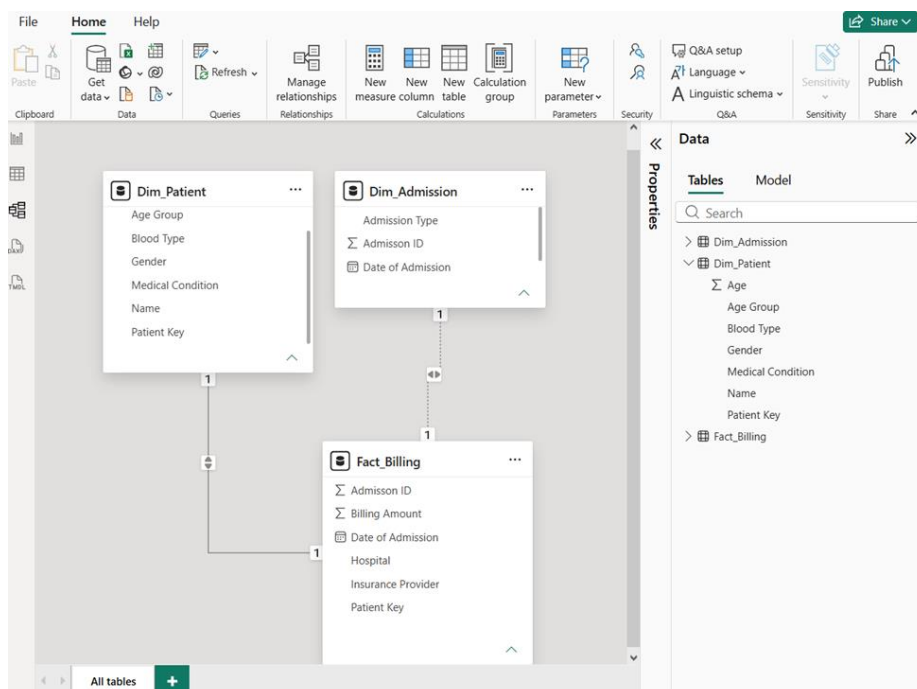
Step 7: Remove the extra raw table

Since I had already split healthcare_dataset into Dim_Patient, Dim_Admission and Fact_Billing, I did not need the original raw table in the model anymore. I right-clicked on healthcare_dataset and chose Delete from model to keep the model clean.



Step 8: Final star-schema model

After removing the raw table, the model was left with just three clean tables: Dim_Patient and Dim_Admission on one side, connected to Fact_Billing in the middle. This is a simple star-schema, with the dimension tables around the fact table.



6. Observations

- The raw data was already fairly clean, so most of the work was in organizing it into separate tables rather than fixing errors.
- Splitting one big table into Dim_Patient, Dim_Admission and Fact_Billing makes the model easier to understand and query.
- The Age Group column makes it much easier to analyze patients by age category instead of using the raw age number.
- Using a Patient Key to join the tables avoids repeating patient details again and again in the fact table.

7. Conclusion

This week I learned how to clean a healthcare dataset in Power Query, create a derived column using conditional logic, and split one big table into proper dimension and fact tables. I also learned how to connect these tables with relationships and remove the extra raw table to keep a clean star-schema model. This kind of model makes it much easier to build an interactive Power BI report for clinical analysis.